

ML Assignment Report 2

AI211 Machine Learning

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Question 1 : Experimental Analysis of Logistic Regression Variants

0.1 Observations

Class Distribution:

From the class distribution plot we can see that the dataset is not perfectly balanced. The *Low* class has more number of samples compared to *Medium* and *High*. The *High* class has the least samples because it is formed by combining *good* and *vgood* categories. Due to this imbalance, the model may sometimes bias more towards predicting the majority class.

Feature Class Relationship:

From the feature vs class bar plots, it can be observed that some features show clear difference between classes while some features show overlap. For example, certain feature values appear more in Low class compared to Medium or High class. However many feature categories are shared between adjacent classes (Low–Medium and Medium–High), so the separation is not very clear.

This indicates that a single feature alone is not enough to classify the car evaluation properly and the model needs to consider multiple features together.

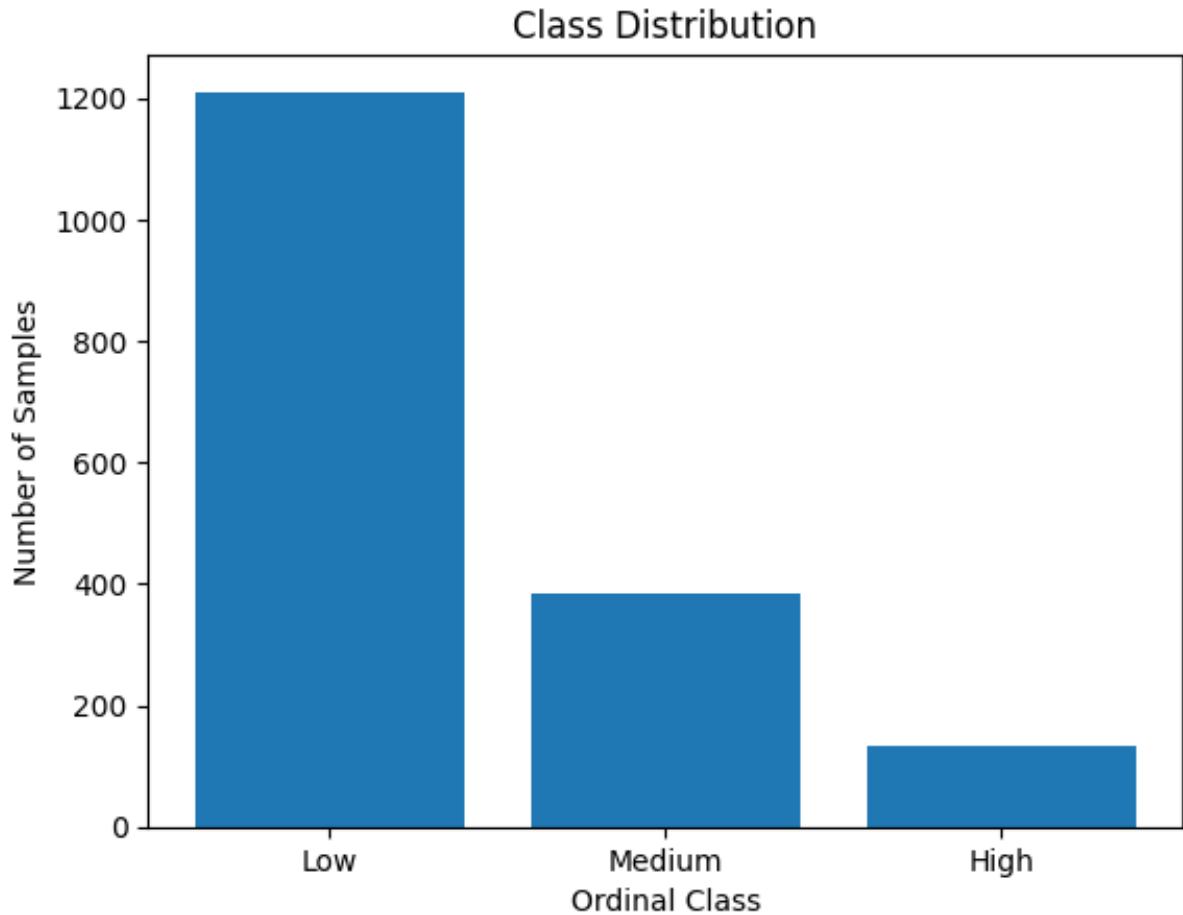


Figure 1: Class Distribution of the Car Evaluation Dataset After Ordinal Label Mapping

Observations from Figure 1:

From the class distribution plot it can be seen that the dataset is not balanced. The *Low* class has very large number of samples compared to the other classes. The *Medium* class has moderate number of samples while the *High* class has the least samples in the dataset.

This shows that there is class imbalance present. Because of this the model may sometimes predict the Low class more frequently. Also since High class samples are less, the model may find it little difficult to learn patterns for that class properly.



Figure 2: Distribution of Safety Feature Across Ordinal Classes

Observations from Figure 2:

- When safety level is *low*, almost all samples belong to the Low class and there are almost no Medium or High class samples.
- For *medium* safety, samples are distributed between Low and Medium classes with few High class samples.
- When safety level is *high*, the number of Medium and High class samples increases.
- This shows that safety feature plays an important role in determining the car evaluation class.

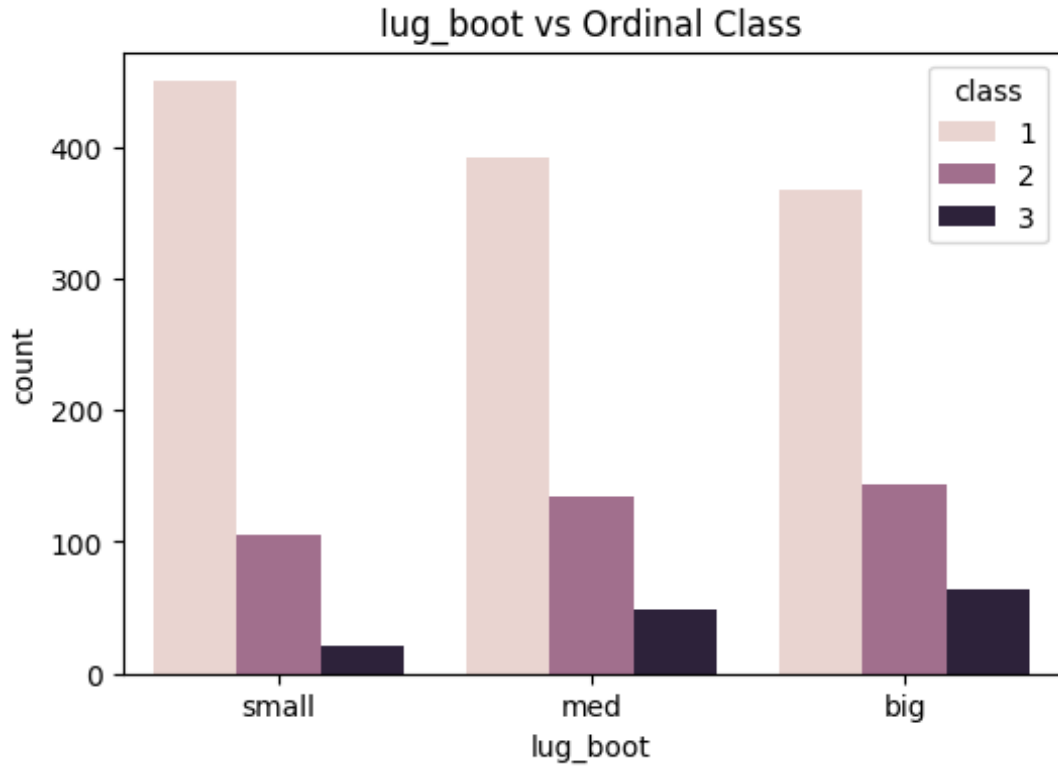


Figure 3: Distribution of Luggage Boot Size Across Ordinal Classes

Observations from Figure 3:

- For all luggage boot sizes (small, medium, big), the Low class has the highest number of samples.
- Medium and High class samples increase slightly as the luggage boot size becomes larger.
- However the difference between classes is not very large, showing that this feature alone does not strongly separate the classes.
- This indicates some overlap between ordinal classes with respect to the `lug_boot` feature.



Figure 4: Distribution of Passenger Capacity Across Ordinal Classes

Observations from Figure 4:

- When the number of persons is 2, almost all samples belong to the Low class.
- When passenger capacity increases to 4 or more, the number of Medium and High class samples increases.
- This shows that higher passenger capacity contributes positively to car acceptability.
- Therefore the persons feature provides some separation between the ordinal classes.

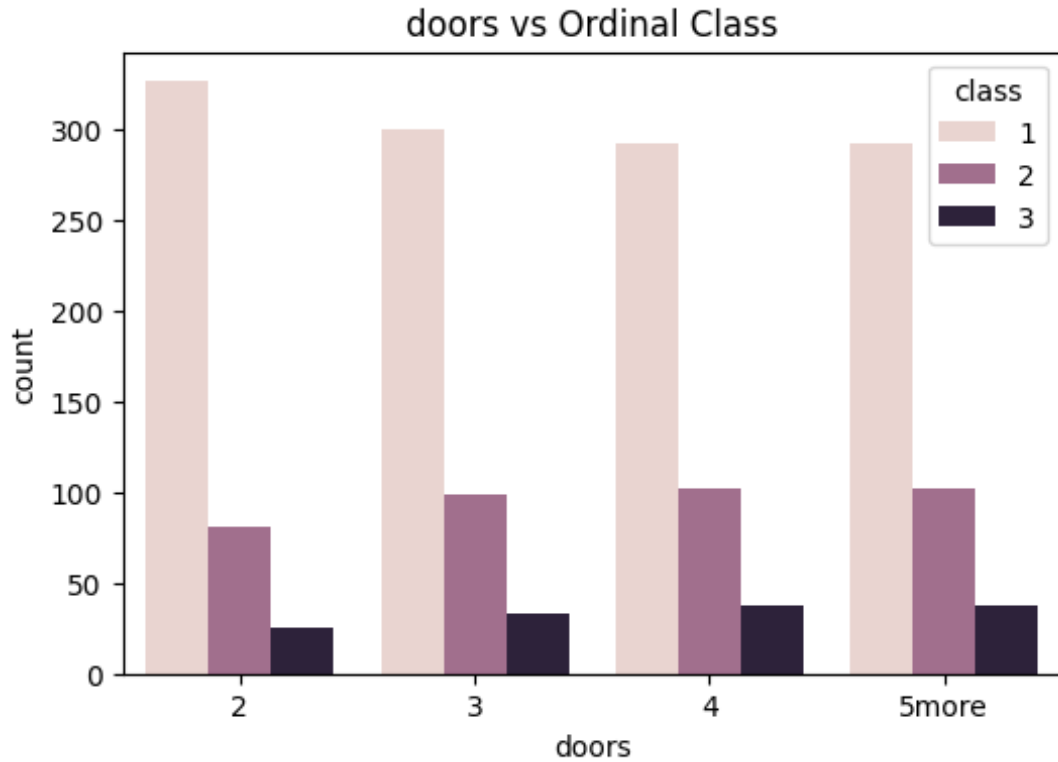


Figure 5: Distribution of Number of Doors Across Ordinal Classes

Observations from Figure 5:

- The Low class has the highest count across all door categories.
- Medium and High class samples are present for all door values but in smaller numbers.
- The distribution pattern across different door counts looks quite similar for each class.
- This suggests that the number of doors does not strongly influence the car evaluation class.

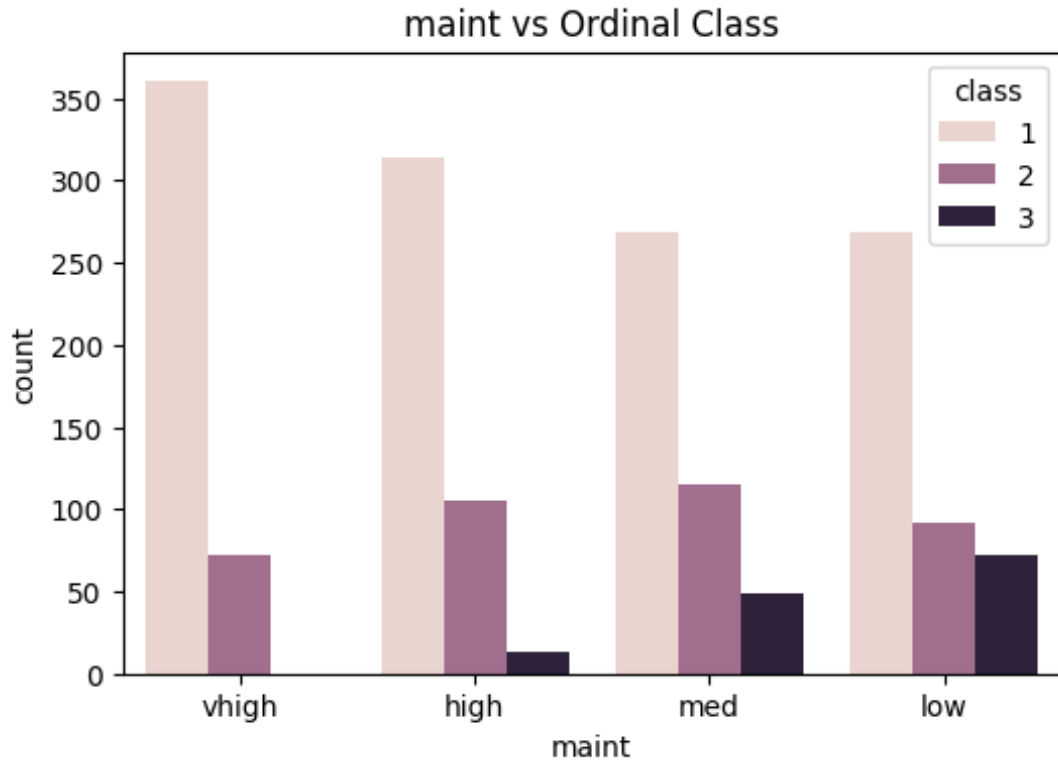


Figure 6: Distribution of Maintenance Cost Across Ordinal Classes

Observations from Figure 6:

- When maintenance cost is very high or high, most samples belong to the Low class.
- As maintenance cost decreases to medium and low, the number of Medium and High class samples increases.
- This indicates that cars with lower maintenance cost are more likely to have better evaluation.
- Hence maintenance cost has noticeable effect on the ordinal classification.

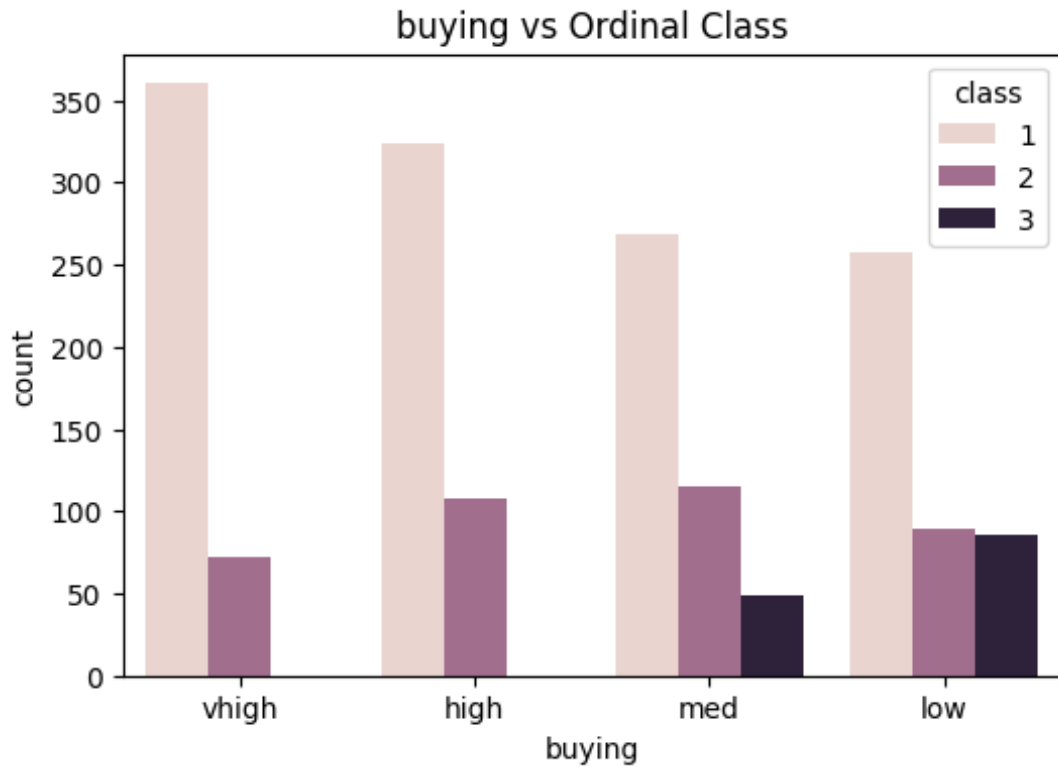


Figure 7: Distribution of Buying Price Across Ordinal Classes

Observations from Figure 7:

- When buying price is very high or high, most samples belong to the Low class.
- As buying price decreases to medium and low, the number of Medium and High class samples increases.
- This suggests that lower buying price makes the car more acceptable.
- However there is still some overlap between classes, meaning buying price alone cannot fully separate the classes.

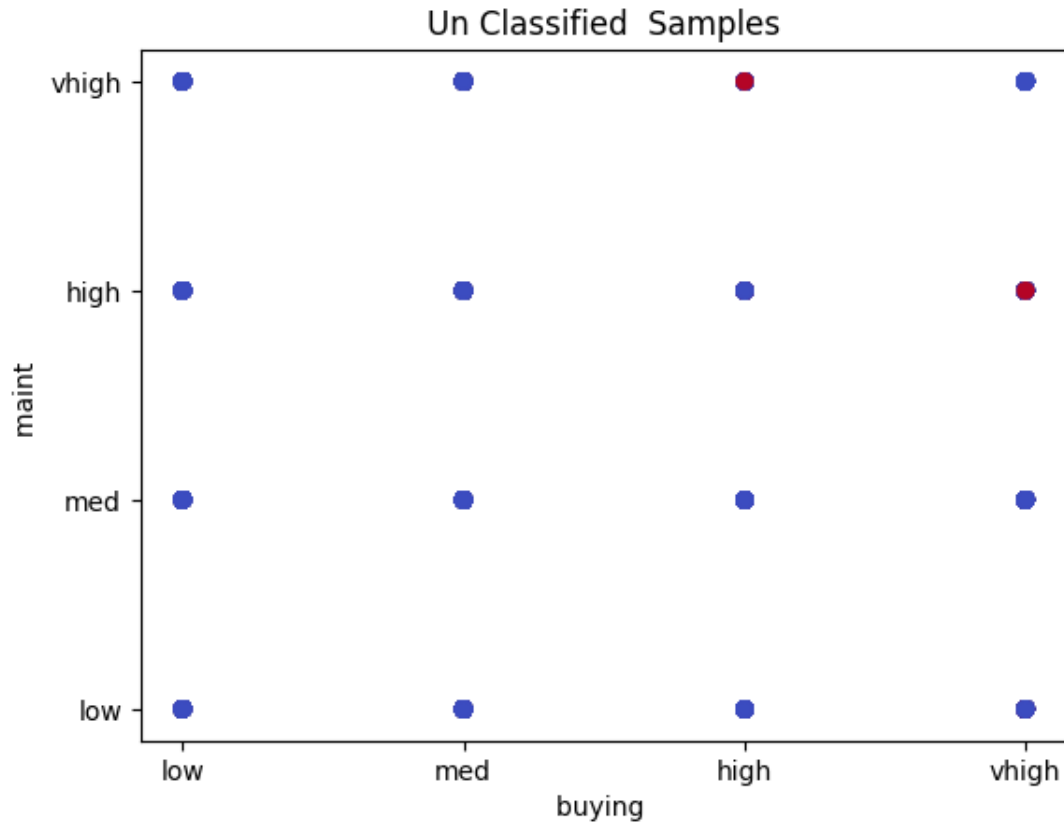


Figure 8: Visualization of Misclassified Samples using Buying and Maintenance Features

Observations from Figure 8:

- The plot shows the classification results of the linear classifier using the *buying* and *maint* features.
- Most of the points are classified correctly, which are shown in blue color.
- A few points are marked in red color which represents the misclassified samples.
- These misclassified points occur mainly due to overlap between different ordinal classes in the feature space.
- Since only two features are used for visualization, the linear classifier cannot perfectly separate all the classes.

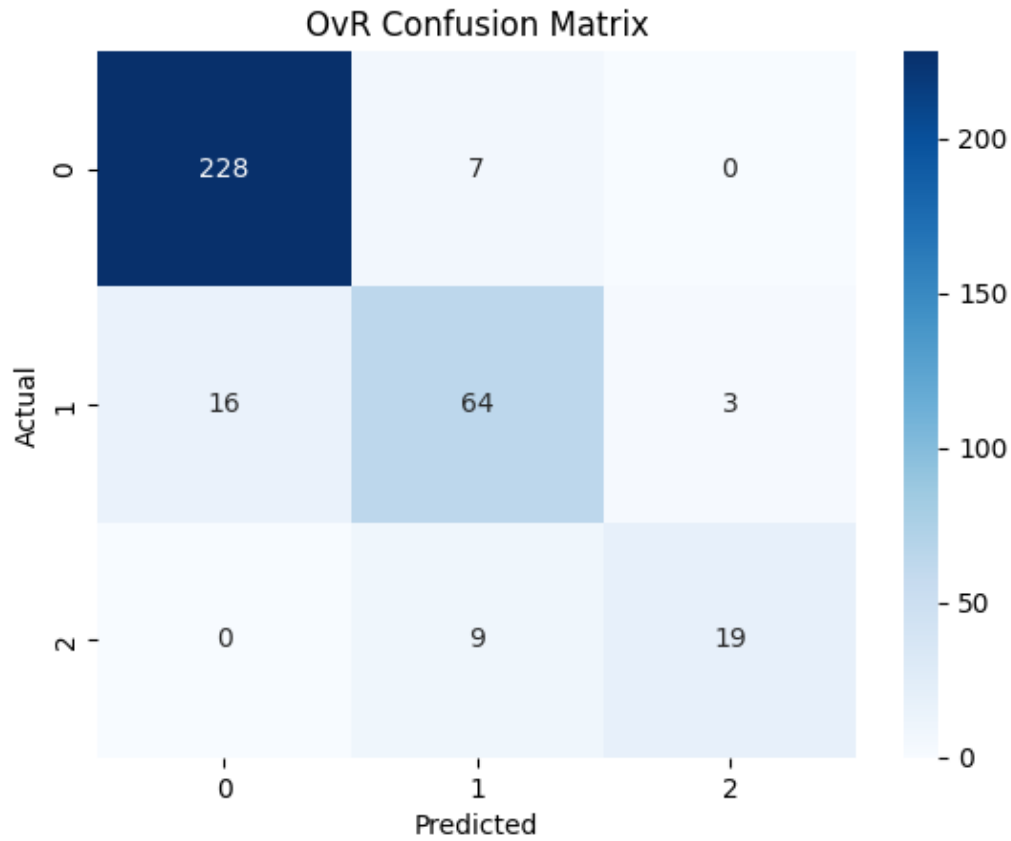


Figure 9: Confusion Matrix for One vs Rest Logistic Regression

Observations from Figure 9:

- Most of the samples lie along the diagonal of the confusion matrix, which indicates that the model correctly classifies majority of the samples.
- The Low class has the highest number of correct predictions compared to the other classes.
- Some Medium class samples are misclassified as Low, which shows overlap between these two classes.
- A small number of High class samples are misclassified as Medium.
- There are almost no extreme misclassifications between Low and High classes, indicating that the model generally respects the ordinal structure of the labels.

1 Observations for Task 2: Decision Tree Classification

Dataset Understanding:

- The Iris dataset contains 150 samples with 4 numerical features which are sepal length, sepal width, petal length and petal width.
- The dataset has three classes which are Setosa, Versicolor and Virginica.
- The dataset was divided into training and testing sets using 80:20 split with random seed value 42 to ensure reproducibility.
- From the dataset we understood how classification problems work with numerical feature values.

Training Decision Tree using Entropy:

- The decision tree classifier was implemented from scratch using entropy as the splitting criterion.
- Entropy measures the impurity of the dataset and the algorithm selects the feature split which gives the highest information gain.
- The tree recursively splits the dataset based on the best feature and threshold until the stopping condition is reached.
- From this experiment we understood how decision trees automatically learn decision rules from the training data.

Entropy vs Gini Index:

- Two decision tree models were trained using entropy and gini index as the splitting criteria.
- Both methods measure the impurity of a node but use different mathematical formulas.
- In most cases the classification performance of entropy and gini index was very similar.
- This shows that both criteria are effective for building decision tree classifiers.

Effect of Maximum Depth:

- The maximum depth of the decision tree controls the complexity of the model.
- When depth is small the tree becomes simple and may underfit the dataset.
- When depth increases the tree can learn more detailed patterns from the training data.
- However very large depth may cause overfitting where the model performs well on training data but poorly on test data.

Random Forest Classifier:

- Random Forest was implemented by training multiple decision trees on different bootstrap samples of the dataset.
- Each tree makes its own prediction and the final output is obtained using majority voting.
- This ensemble approach reduces variance and improves stability of predictions.
- Compared to a single decision tree, the Random Forest model usually gives better generalization performance.

Question 3 : Task 3 (Support Vector Machines)

OVO Linear SVM Implementation

In this task we implemented a One vs One (OVO) linear SVM classifier. For a multi class problem with three classes, separate binary classifiers are trained for each pair of classes. Each classifier learns a separating hyperplane that maximizes the margin between the two classes while allowing some violations controlled by the penalty parameter C .

Role of Penalty Parameter C

The parameter C controls the trade off between maximizing the margin and minimizing classification errors.

- Small values of C allow a wider margin but tolerate more misclassification.
- Large values of C penalize misclassification strongly and lead to narrower margins.
- Proper tuning of C is important to balance model complexity and generalization.

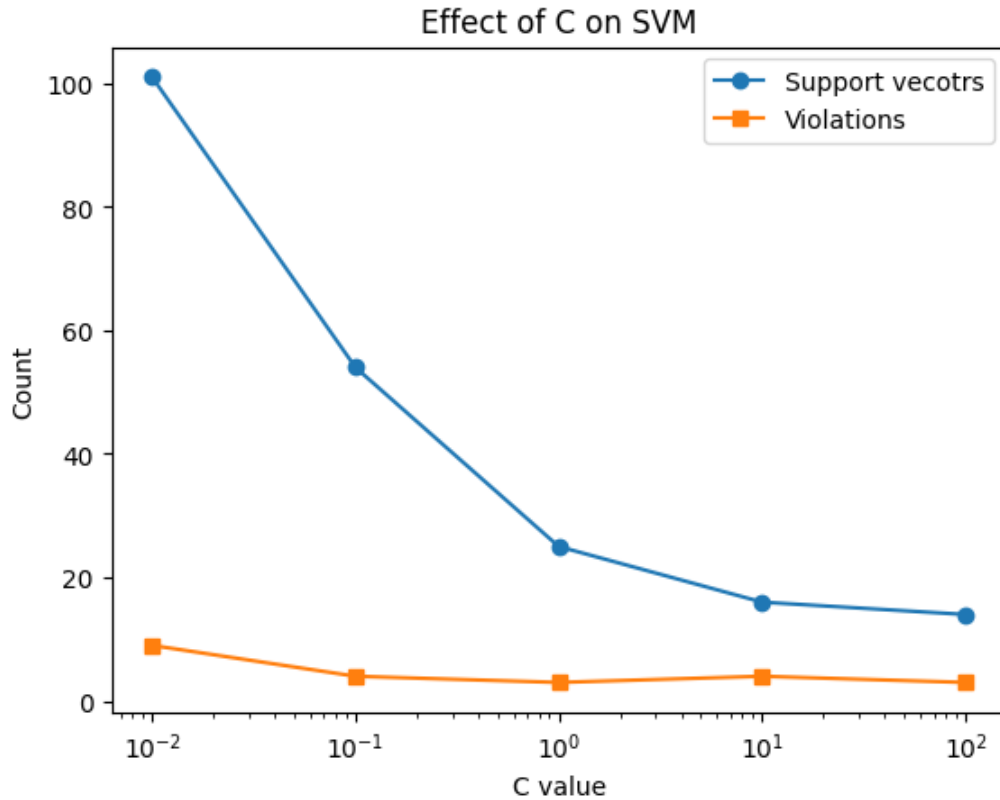


Figure 10: Effect of the penalty parameter C on the number of support vectors and violated points

Observations from Figure 10:

- When the value of C is very small, the number of support vectors is very high because the model allows a large margin and many points lie inside the margin region.
- As the value of C increases, the number of support vectors decreases since the model tries to fit the training data more strictly.
- The number of violated points (points inside the margin or misclassified) reduces slightly as C increases.
- For very large values of C , the margin becomes narrower and the classifier focuses more on minimizing training error.
- This demonstrates the trade off between margin size and classification accuracy controlled by the parameter C .

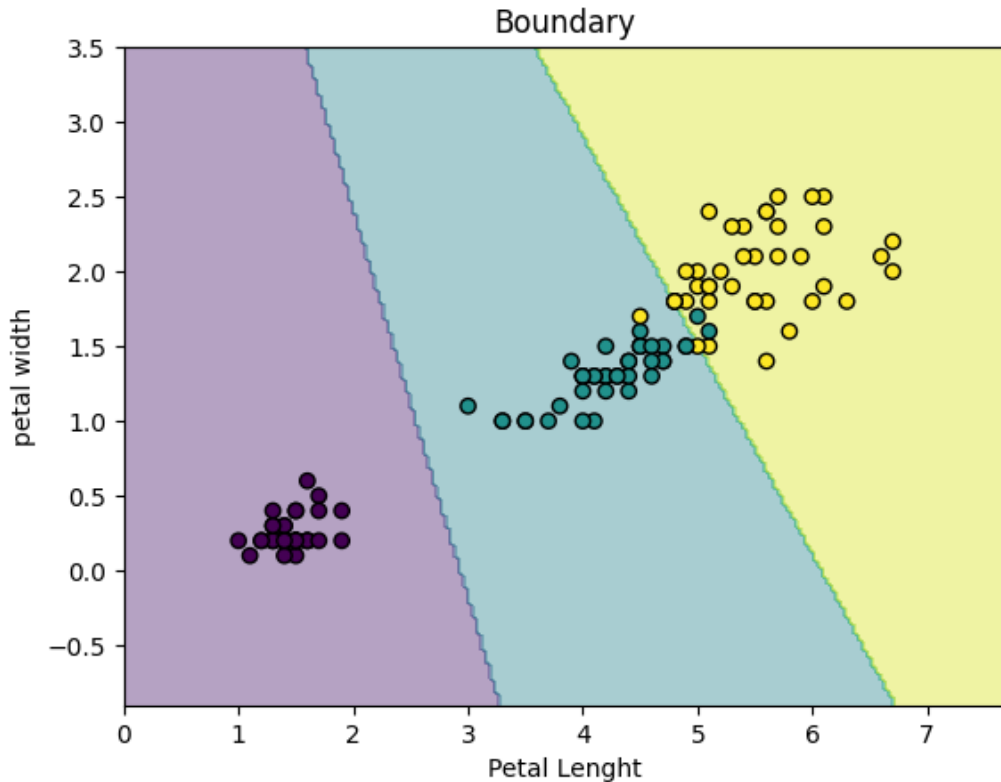


Figure 11: Decision boundary of the linear SVM using petal length and petal width

Observations from Figure 11:

- The plot shows the decision boundaries learned by the linear SVM using petal length and petal width as features.
- The data points form three clusters corresponding to the three iris species.
- The linear decision boundaries separate the classes reasonably well, especially the Setosa class which is clearly separable from the other two classes.
- Some overlap is observed between Versicolor and Virginica classes, which makes the classification slightly difficult.
- The SVM finds hyperplanes that maximize the margin between classes, resulting in clear and structured decision regions.

Kernel SVM Analysis

In addition to linear SVM, polynomial and RBF kernel SVMs were also explored.

- Kernel functions allow the SVM to map the input data into higher dimensional feature spaces.
- The polynomial kernel creates curved decision boundaries which can capture more complex relationships in the data.
- The RBF kernel produces smoother nonlinear boundaries and can adapt to complex structures in the dataset.

- However very high values of kernel parameters may lead to overfitting where the model fits the training data too closely.

Evaluation Metrics

The performance of the best SVM model was evaluated using precision, recall and F1 score.

- Precision measures how many predicted positive samples are actually correct.
- Recall measures how many actual positive samples are correctly identified by the model.
- The F1 score provides a balanced measure combining both precision and recall.
- These metrics help in understanding the overall classification performance of the SVM model.